StatusBoost User Guide

Version 2.0.7

Table of Contents

Preface

1. Purpose
2. Who should read this guide?
3. Where to obtain the latest version of this guide?
4. Providing feedback for this guide

1. Introduction

1.1 What is StatusBoost?

1.1.1 Intended Uses

1.1.2 Core Features

1.1.3 Free Software

1.1.4 Privacy

1.2 System Requirements

1.3 What's New in Version 2.0.7

1.4 What's New in Version 2.0

2. Installing StatusBoost

2.1 Introduction

2.2 Getting the Installer

2.3 Installing StatusBoost

2.4 Upgrading StatusBoost

2.5 Uninstalling StatusBoost

3. User Interface

3.1 Introduction

3.2 Starting Up StatusBoost

3.3 The Main Window

3.4 Default Menu

3.4.1 File Menu

3.4.2 Run Menu

3.4.3 Tools Menu

3.4.4 Help Menu

3.5 Factory Profile Menu

3.5.1 File Menu

3.5.2 Run Menu

3.5.3 Factory Menu

3.5.4 Tools Menu

3.5.5 Help Menu

3.6 Toolbar

3.7 Tabs

3.8 Status Field

3.9 Data Fields

3.10 Notification Message Box

4. Running StatusBoost
4.1 Introduction
4.2 Procedure to Run Software
4.3 Setting Up a Stacked System
4.4 Connecting to a Standalone STS Controller
4.5 Monitoring Several Systems
4.6 Monitoring a UC Alarm Card (StatusBoost .NET Only)
5. Tools Menu Selections
5.1 Introduction
5.2 Device Connection
5.2.1 Find Device
5.3 Network Feature Setup
5.4 System Configuration
5.5 Energy Calibration (Factory Profile Menu Only)
5.6 AC/DC Configuration (Factory Profile Menu Only)
5.7 Rectifier Settings (Factory Profile Menu Only)
6. FAQs
6.1 Introduction
6.2 General Questions
6.3 Software Settings
7. Troubleshooting
7.1 Introduction
7.2 Device Mismatch
7.3 A Stacked System Is Not Built
A. File and Folder Specification
B. StatusBoost .NET

Preface

1. Purpose

StatusBoost is an application that remotely displays the status of Exeltech inverter systems. This guide explains the software's features and the steps to follow to view the status of your inverter system.

2. Who should read this guide?

This guide is for anyone who wants to install StatusBoost to view the current status of their inverter system. It shows how to:

- set up the software;
- run it;
- use the basic menus;
- use the more advanced menus to configure the system.

3. Where to obtain the latest version of this guide?

The guide is included with the StatusBoost software. For the latest version, download the latest StatusBoost from https://exeltech.com/software/.

4. Providing feedback for this guide

If you have questions or suggestions about this guide, send your feedback through https://exeltech.com/contact-us/technical-support/.

1. Introduction

1.1 What is StatusBoost?

StatusBoost is a Java application that remotely displays the status of Exeltech inverter systems on a PC monitor. It reads the system over the local network.

It does not use secure protocols to communicate with the inverter system; it is a quick view of your system's status.

StatusBoost comes as a Windows installer that includes the Java it needs. Section 2, "Installing StatusBoost", describes how to install it. A second edition, StatusBoost .NET, works the same way; Appendix B lists its differences.

1.1.1 Intended Uses

End users can use StatusBoost to confirm that their system is running correctly, without errors. They can also use it to configure their system.

1.1.2 Core Features

- Connects to the inverter system over the local network.
- Reads live data from the system and displays it on the screen.
- Shows a system made of several devices (a **stacked system**) in one tab: up to 5 KC cages and an STS controller.
- Shows a standalone STS controller: its transfer switch status, alarms and phase readings (Section 4.4).
- StatusBoost .NET also shows UC (LCNC) alarm cards, on their own or stacked with an STS controller (Section 4.6).
- Monitors up to 6 systems at once, each in its own tab (Section 4.5).
- Shows when the connection to the system is lost, and reconnects by itself.
- Checks that each configuration change has reached the system before it reports success.
- Saves its settings to an INI file.

1.1.3 Free Software

StatusBoost is free to use, copy and distribute, with no fees. The StatusBoost License Agreement sets out the terms: the installer shows it before installing, and it is published with each release.

1.1.4 Privacy

Exeltech does not collect any information through StatusBoost. It sends nothing about you, your PC, your systems or how you use it to Exeltech or anyone else: there are no analytics, no telemetry, no update checks and no accounts.

StatusBoost talks only to the Exeltech systems on your network. It finds them with a broadcast on the local network (UDP port 6006), reads and configures them with Modbus TCP (port 502), and changes their network settings in Network Feature Setup with SNMP (UDP port 161). What it reads stays on your PC, and its settings are stored only on your PC, in "%APPDATA%\StatusBoost". The links in the Help menu and the About box open Exeltech's website in your browser, and only when you click them.

1.2 System Requirements

- Windows 10 or 11, 64-bit.
- Administrator rights to install StatusBoost (Section 2.3). Java is included; you don't need to install it.
- A network connection to the inverter system. StatusBoost reads the system on TCP port 502, and Find Device searches with UDP port 6006.

1.3 What's New in Version 2.0.7

- **UC alarm cards, in StatusBoost .NET:** StatusBoost .NET shows UC (LCNC) alarm cards, on their own or stacked with an STS controller (Section 4.6). StatusBoost, the Java edition, does not.
- **AB Bus thresholds:** in AC/DC Configuration, the threshold list for 108 V and 120 V systems shows the right voltages (Section 5.6). Only the list's labels changed; the setting sent to the system is the same.

Also new since version 2.0.2, in versions 2.0.3 to 2.0.6, which were not published:

- **The configuration windows send only what you change.** A setting you did not change keeps the system's value, even if the window showed an older one. If you changed nothing, the window says "No changes detected." and sends nothing (Section 5.1).
- **Rectifier Settings** sends the target DC voltage only when you change it, and "Actual" shows the system's values (Section 5.7).
- **Toggle Reference** writes only to a KC system (Section 3.5.3).
- **Gentler on the system's network card:** StatusBoost opens one connection at a time to a system, a short moment apart.
- **Several systems, a tab each:** "New Tab" in the "File" menu (Ctrl+T) opens a tab for another system. StatusBoost reads up to 6 systems at once (Section 4.5).
- **Tab headers:** each tab shows a status dot (green while running, yellow for an alarm, red when stopped or not reached), the system's name, and a cross that deletes the tab (Section 3.7).
- **Run All and Stop All** in the "Run" menu start and stop every tab. The other menus and the toolbar act on the selected tab.
- **One device, one tab:** Find Device leaves out the devices that other tabs already show, and says how many it left out.
- **Your tabs come back:** StatusBoost reopens the tabs of the last session, with the tab you had selected.
- **The window appears at once** when StatusBoost starts, and fills in as each system is read.
- **The transfer switch's notice no longer makes the window taller;** the tab scrolls instead.
- **Standalone STS controllers:** StatusBoost shows an STS controller on its own: its transfer switch status, alarms and the readings of each phase (Section 4.4). It also reads the STS controller of a stacked system.

- **Find Device** searches when you click "Refresh", not as soon as it opens. To find STS controllers, check "Include legacy devices" first (Section 5.2.1).
- **Menus of a stacked system:** Network Feature Setup, System Configuration and the factory items set up one networked device, so a stacked system hides them (Section 4.3).
- **Factory profile:** "Toggle Reference" replaces "Initiate Move to Reference 2" (Section 3.5.3).

1.4 What's New in Version 2.0

Compared with StatusBoost 1.03:

- **Stacked systems:** combine several devices into one system in Find Device (Section 4.3). Each further cage is shown with its own power modules and alarms.
- **Connection loss:** when the system stops answering, StatusBoost shows "Disconnected" and blank (N/A) fields, and reconnects by itself (Section 3.8).
- **AB Bus:** the field shows only when AB-bus monitoring is enabled. It shows "Fail" when the system reports an AB-bus fault.
- **Transfer switches:** KC Skinny transfer switches show the transfer switch fields too.
- **Safer configuration:** each change is sent separately and confirmed by the system (Section 5.1).
- **Network Feature Setup:** "Update" no longer turns off the system's SNMP agent and web server. It checks the addresses, asks before restarting the network card, sends only what changed and never a setting it could not read, and restarts the card only when everything was sent (Section 5.3).
- **Installer:** StatusBoost now comes as a Windows installer that includes its own Java, so you no longer install Java. It adds StatusBoost to the Start menu and the desktop (Section 2).
- **Settings folder:** your settings are kept in "%APPDATA%\StatusBoost", so upgrading or uninstalling StatusBoost keeps them (Section 6.3).

2. Installing StatusBoost

2.1 Introduction

This section describes how to install, upgrade and uninstall StatusBoost on your PC.

StatusBoost comes as a Windows installer, "StatusBoost-<version>.msi", for example "StatusBoost-2.0.7.msi". The installer includes the Java that StatusBoost needs, so you don't have to install Java.

2.2 Getting the Installer

Download the installer from https://exeltech.com/kc-inverter-software/, or copy it from the CD shipped with your system.

2.3 Installing StatusBoost

You need administrator rights on the PC to install StatusBoost.

1. Double-click "StatusBoost-<version>.msi".
2. If Windows shows "Windows protected your PC", click "More info", then "Run anyway". A signed installer shows Exeltech as the publisher there; the test versions are not signed yet, and show "Unknown publisher".
3. On the welcome page, click "Next". Read the license agreement, accept it, and click "Next".
4. Choose the folder, or keep the suggested "C:\Program Files\StatusBoost", and click "Next", then "Install".
5. When Windows asks whether the installer may make changes to your PC, click "Yes".
6. Click "Finish".

The installer adds StatusBoost to the Start menu, in the "Exeltech" folder, and puts a StatusBoost shortcut on the desktop. It also adds a Windows Firewall rule, "StatusBoost", that lets StatusBoost receive the systems' answers from your local network; uninstalling StatusBoost removes it. Section 3.2 explains how to start it.

2.4 Upgrading StatusBoost

To install a newer version, run its installer as in Section 2.3. It replaces the installed version, and your settings are kept, because StatusBoost keeps them in "%APPDATA%\StatusBoost", not in its installation folder (Section 6.3).

If you used StatusBoost 1.x, or a StatusBoost 2.0 folder copied from a zip file or CD, its settings are in "statb.ini" in that StatusBoost folder. To keep your systems, do this before you first start the installed StatusBoost:

1. In File Explorer, type `%APPDATA%` in the address bar and press Enter.
2. Make a folder named "StatusBoost" there, unless there is one.

3. Copy "statb.ini" from your old StatusBoost folder into it.

If you skip this, choose your system again with "Device Connection" (Section 4.2). Once your systems show as before, you can delete the old StatusBoost folder.

2.5 Uninstalling StatusBoost

1. Click the "Start" button, then "Settings", then "Apps".
2. Find "StatusBoost" in the list, and click "Uninstall".
3. When Windows asks whether the installer may make changes to your PC, click "Yes". If you don't, the uninstall stops with the message "You must be an Administrator to remove this application".

Your settings stay in "%APPDATA%\StatusBoost", so a later installation finds them again. To remove them too, delete that folder.

3. User Interface

3.1 Introduction

This section introduces the main window, to help you get acquainted with StatusBoost.

3.2 Starting Up StatusBoost

Start StatusBoost from the Start menu (in the "Exeltech" folder) or with the StatusBoost shortcut on the desktop.

The main window opens at once. While the first tab has no system yet, for example the first time, an error message appears on top of it (Figure 3.1). Click "OK". Section 4.2, "Procedure to Run Software", explains how to choose your system.

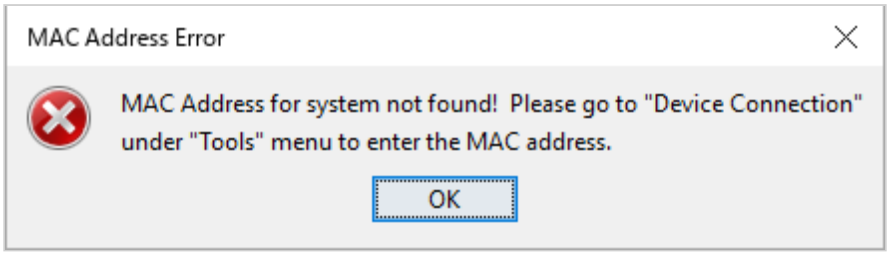


Figure 3.1. MAC Address Error

3.3 The Main Window

The main window shows one tab for each system (Figure 3.2).

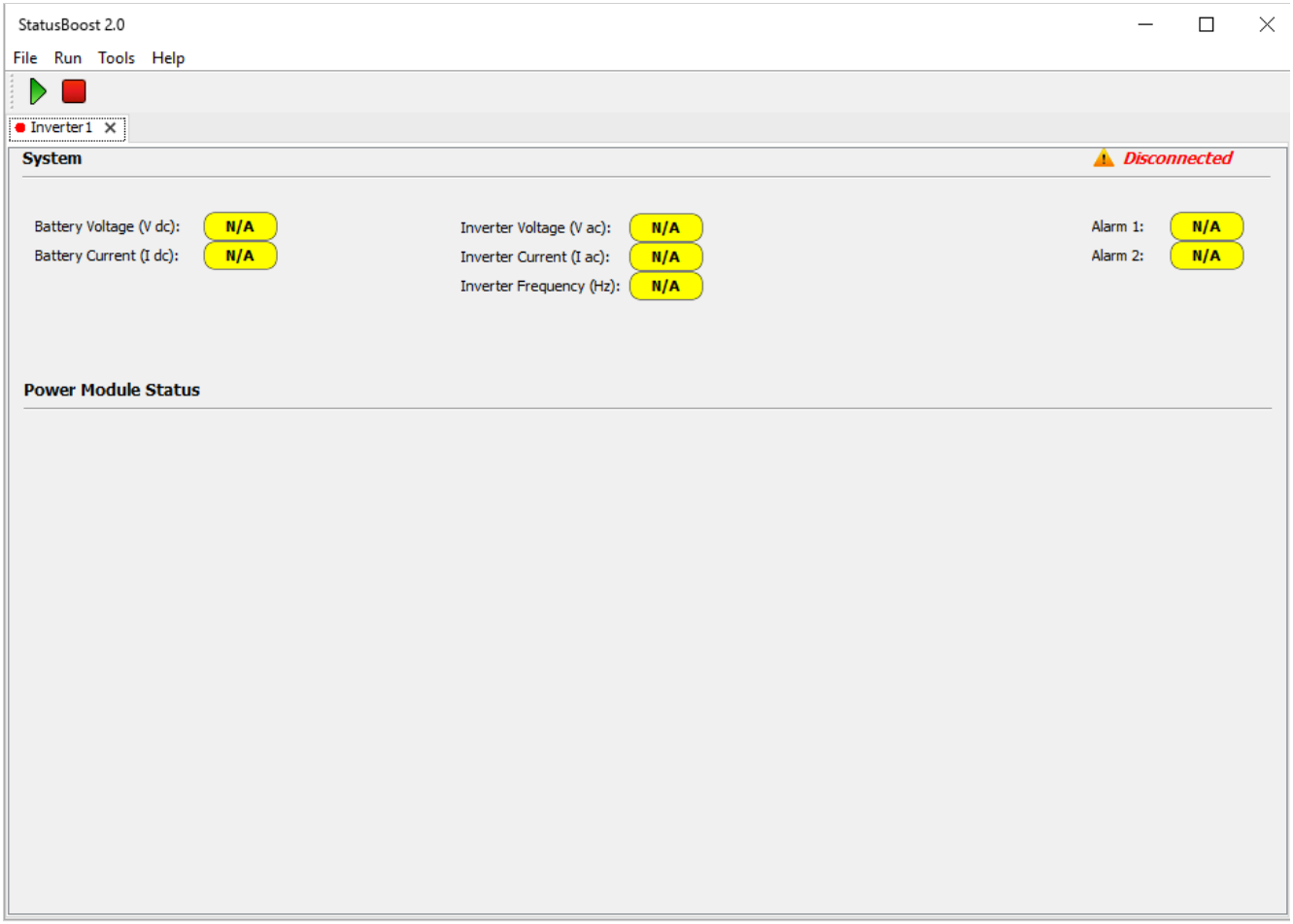


Figure 3.2. Main Window Example

The main window has these features:

1. **The menu:** the list of commands at the top of the window, right under the title bar. Each menu opens further commands.
2. **The toolbar:** the small area under the menu with buttons for frequently used menu items.
3. **The tabs:** one for each system StatusBoost monitors. Each shows a status dot, the system's name and a cross that deletes the tab (Section 3.7).
4. **The tabbed pane:** the area under the tabs. It shows the data fields of the selected tab's system and its power modules. When they don't fit in the window, for example for a stacked system of several cages, scroll down to see the rest.
5. **The status field:** in the upper-right corner of the tabbed pane. It shows whether StatusBoost is running.
6. **The data fields:** show the information read from the system. Fields with no data show "N/A".
7. **The notification message box:** not shown in Figure 3.2. It appears near the top of the tabbed pane when something needs your attention.

3.4 Default Menu

StatusBoost's default menu has four main menu items (Figure 3.3).



Figure 3.3. Default Menu

Menu	Purpose
File	Add and delete tabs, save the settings to the INI file, and close the software.
Run	Start ("Run") and stop StatusBoost, for the selected tab or for every tab.
Tools	Choose the system to connect to; set network features over SNMP; set alarm, module slot and cage configurations for the system.
Help	Open the user guide and the company website links, and see version information.

The menus act on the system of the selected tab (Section 4.5). After you choose a different system, some menu items are unavailable (greyed out) until StatusBoost has read the new system. A stacked system hides Network Feature Setup and System Configuration (Section 4.3). With a standalone STS controller they are unavailable (Section 4.4).

3.4.1 File Menu

The "File" menu items are shown in Figure 3.4 and Table 3.1.

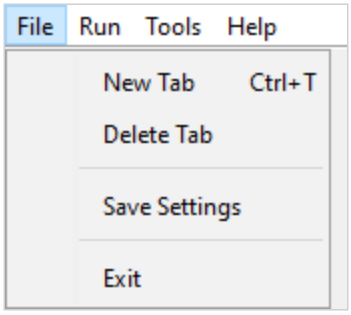


Figure 3.4. File Menu

Table 3.1

Menu Item	Description
New Tab	Opens a tab for another system, with the Device Connection window (Section 4.5). Shortcut: Ctrl+T. Unavailable while 6 tabs are open.
Delete Tab	Deletes the selected tab: StatusBoost asks first, stops reading its system and removes it from the INI file (Section 4.5).
Save Settings	Saves the settings to the INI file. StatusBoost usually saves its settings by itself, so this item is seldom needed. The INI file, "statb.ini", holds the devices of every tab (MAC address, Modbus key, serial number, configuration, name, description, phase count), the selected tab, the factory profile flag and the logs folder.

Menu Item	Description
Exit	Closes StatusBoost.

3.4.2 Run Menu

The "Run" menu items are shown in Figure 3.5 and Table 3.2.

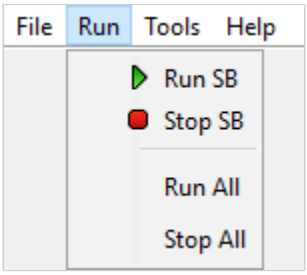


Figure 3.5. Run Menu

Table 3.2

Menu Item	Description
Run SB	Connects to the selected tab's system, reads its data and displays it. In a stacked system, every device is read.
Stop SB	Stops reading the selected tab's system and clears its data from the screen.
Run All	Runs every tab, as "Run SB" does for one.
Stop All	Stops every tab.

3.4.3 Tools Menu

The "Tools" menu items are shown in Figure 3.6 and Table 3.3.

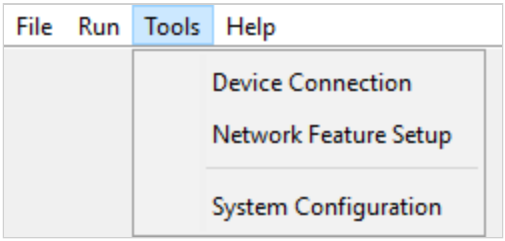


Figure 3.6. Tools Menu

Table 3.3

Menu Item	Description
Device Connection	Choose the system to connect to: enter its MAC address, or find it on the network. You can also combine several devices into a stacked system, and enter a name and description for the system.
Network Feature Setup	Set the system's IP address type (DHCP or static), trap IP addresses, keep-alive trap interval, SNTP server IP address and local time zone, over the SNMP protocol. Not shown for a stacked system.
System Configuration	Set the alarm conditions, and which module slots and cages the system has. Not shown for a stacked system.

3.4.4 Help Menu

The "Help" menu items are shown in Figure 3.7 and Table 3.4.

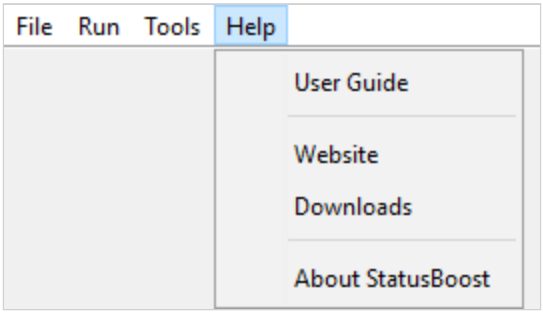


Figure 3.7. Help Menu

Table 3.4

Menu Item	Description
User Guide	Opens this guide ("sbguide.pdf" in the "app" folder of the StatusBoost installation, Appendix A).
Website	Opens the home page of the company website.
Downloads	Opens the software download page of the company website.
About StatusBoost	Shows version information: StatusBoost's version, the Java it runs on, and the connected system's phase and gateway firmware builds (Figure 3.8).

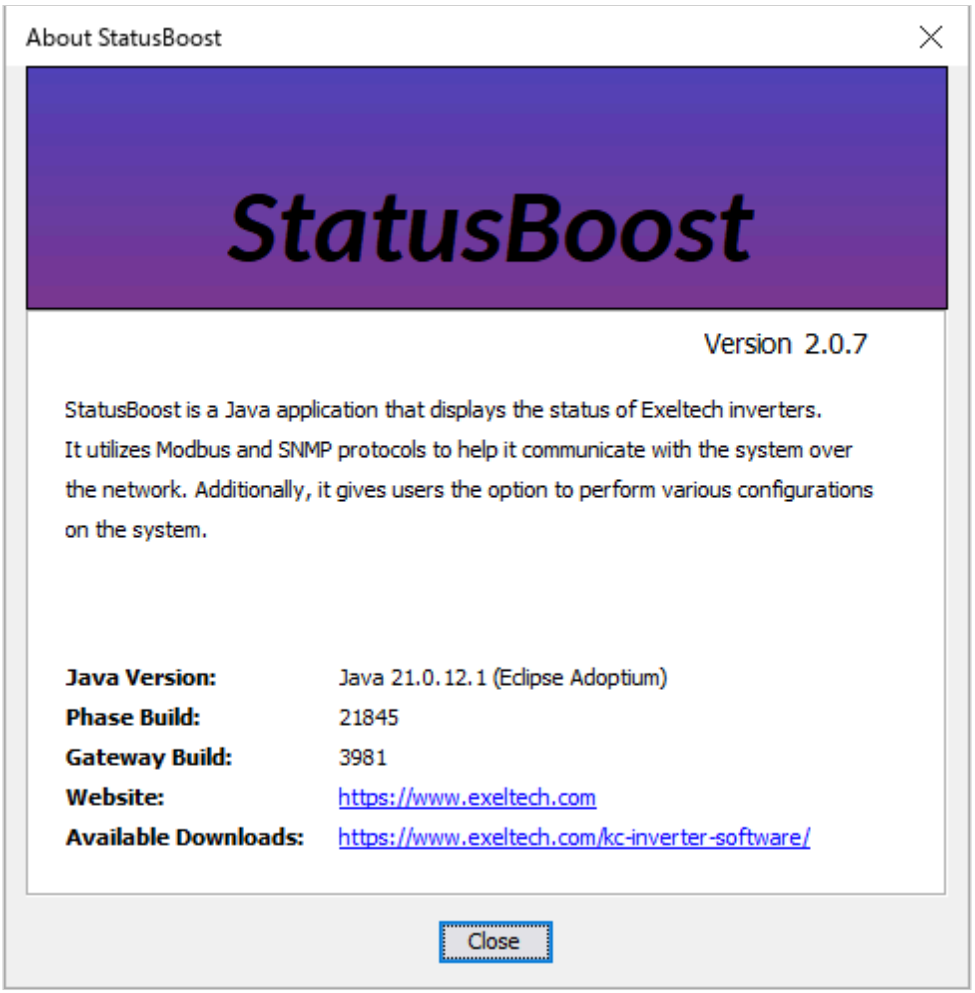


Figure 3.8. About StatusBoost

3.5 Factory Profile Menu

The factory profile menu is available only when StatusBoost is set up to run with the factory profile. To use it, set the "FactoryProfile" parameter in "statb.ini" (in "%APPDATA%\StatusBoost", Section 6.3) to "Y" before starting StatusBoost. The factory profile menu has five main menu items (Figure 3.9).

In the factory profile, the Rectifier Settings window (Section 5.7) also opens when StatusBoost starts, for the selected tab, unless its system is a stacked system. With several tabs, its title starts with the system's name.



Figure 3.9. Factory Profile Menu

Menu	Purpose
File	Same as in the default menu (Section 3.4).
Run	Same as in the default menu.
Factory	Setup steps for factory personnel before delivery.

Menu	Purpose
Tools	The default Tools items, plus the AC/DC voltage and current settings, energy calibration and rectifier settings.
Help	Same as in the default menu.

3.5.1 File Menu

Same as in Section 3.4, "Default Menu".

3.5.2 Run Menu

Same as in Section 3.4, "Default Menu".

3.5.3 Factory Menu

This menu holds the procedures factory personnel use to set up systems before delivery (Figure 3.10).

"Toggle Reference" writes the system's reference coil (coil 0): 1 the first time, then 0, then 1 again, and so on. Each time, StatusBoost says which value it wrote. If a write fails, the next "Toggle Reference" tries the same value again. Connect to the system first (Section 4.2). StatusBoost reads the system's type first and writes only to a KC system, or in StatusBoost .NET also to a UC alarm card (Section 4.6).

"Assign System Model & Serial" sets a unit's model and serial number from its two barcodes. Scan the unit's MAC address barcode into "MAC Address" and its system barcode into "System Barcode", then click "Set". StatusBoost finds the unit on the network by that MAC address, sets the model number, then the serial number, and shows whether it worked, for example "Set on 192.168.1.20.". It does not depend on the system shown in the tab.

A stacked system hides "Toggle Reference"; the Factory menu then shows only "Assign System Model & Serial" (Section 4.3).

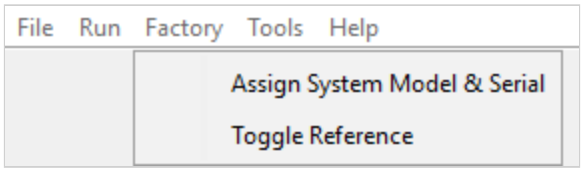


Figure 3.10. Factory Menu

3.5.4 Tools Menu

The factory profile's "Tools" menu has the default items plus "Energy Calibration", "AC/DC Configuration" and "Rectifier Settings" (Figure 3.11, Table 3.5). A stacked system hides all of them except "Device Connection" (Section 4.3).

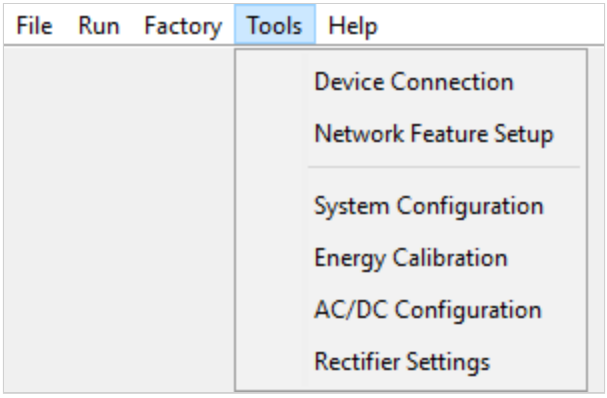


Figure 3.11. Tools Menu (Factory Profile)

Table 3.5

Menu Item	Description
Device Connection	Same as in Section 3.4, "Default Menu".
Network Feature Setup	Same as in Section 3.4.
System Configuration	Same as in Section 3.4.
Energy Calibration	Calibrates the system's phase voltages and currents.
AC/DC Configuration	Sets the system's AC and DC voltage and current parameters and the AB-bus options.
Rectifier Settings	Shows and sets the rectifier's target DC voltage and maximum magnitude.

3.5.5 Help Menu

Same as in Section 3.4, "Default Menu".

3.6 Toolbar

The toolbar gives quick access to frequently used menu items (Figure 3.12, Table 3.6).



Figure 3.12. Toolbar

Table 3.6

Toolbar Button	Menu Item	Description
Green triangle	Run → Run SB	Connects to the selected tab's system, reads its data and displays it. The button stays pressed while that tab runs.

Toolbar Button	Menu Item	Description
Red square	Run → Stop SB	Stops reading the selected tab's system and clears its data from the screen.

3.7 Tabs

Each tab shows one system (Figure 3.13; Section 4.5). A tab's header has:

- **a status dot:** green while StatusBoost reads the system, yellow while one of its cages, or a standalone STS controller, reports an alarm (Alarm 1 or Alarm 2), and red while it is stopped or can't be reached;
- **the system's name,** the "Inverter Name" of the Device Connection window (Section 5.2):
- a device chosen with "Find..." is first named after its MAC address, and a stacked system "Stacked System"; change the name there if you like;
- a system without a name, or a tab without a system yet, is called "Inverter1";
- while a new tab's Device Connection window is open, the tab is called "New Tab";
- **a cross,** which deletes the tab (Section 4.5).

If you entered a description, it appears as a tooltip when you hover the mouse over the name.

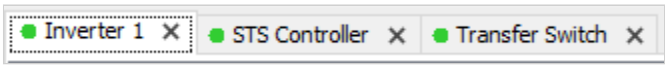


Figure 3.13. Tabs with Their Status, Names and Delete Buttons

3.8 Status Field

The status field, in the upper-right corner of the tabbed pane, shows one of three states for the selected tab's system (Figure 3.14):

- **Disconnected:**
- No system has been set up in the Device Connection window yet; or
- StatusBoost cannot reach the system, for example after a network or power problem. All fields then show "N/A". StatusBoost keeps trying every few seconds and shows "Running" again as soon as the system answers.
- **Stopped:** a system has been set up, but StatusBoost is not running.
- **Running:** StatusBoost is reading the system. Start it with the green triangle on the toolbar, or "Run SB" in the "Run" menu.



Figure 3.14. Status Field: Disconnected, Stopped, Running

3.9 Data Fields

The tabbed pane shows groups of data fields, depending on your system:

- **System:** always.
- **Transfer Switch Status,** with the utility and output values: for a system with a transfer switch.
- **Power Module Status:** always.
- **Power Module Status Cage 2, 3, ...:** for each further cage of a stacked system.



Figure 3.15. Data Fields Example 1 (Monitor Card Only System)



Figure 3.16. Data Fields Example 2 (Monitor Card and Transfer Switch Combo System)



Figure 3.17. Data Fields Example 3 (Stacked System of Two Cages)

System fields

Field	Description
Battery Voltage	The input voltage of the system. It should be close to the nominal DC voltage of your system: 24, 48, 108 or 120 VDC.
Battery Current	The input current of the system. In a stacked system, the total of all cages.
Inverter Voltage	The output voltage. It should be close to the nominal AC voltage, 120 VAC.
Inverter Current	The output current.
Inverter Frequency	The output frequency. It should be close to 60 Hz.
Alarm 1 (Minor level)	"Good" while the system runs within acceptable parameters. "Warning" when an Alarm 1 condition checked in System Configuration occurs. In a stacked system it appears in Power Module Status Cage 1 instead.

Field	Description
Alarm 2 (Major level)	"Good" while the system runs within acceptable parameters. "Fail" when an Alarm 2 condition checked in System Configuration occurs. In a stacked system it appears in Power Module Status Cage 1 instead.
AB Bus	Shown only when AB-bus monitoring is enabled for the system (AC/DC Configuration, AB Bus Status "On"). "Good" while the AB bus is fine, "Fail" when the system reports an AB-bus fault.

Power module fields

Field	Description
Slot N	One field per power module, where N is the slot number. Only the slots in use are shown, side by side. "Online": the module is working. "Attention": it reports a problem. "Offline": it is not responding. "N/A": the slot is checked in System Configuration but no module was found there. Double-click a yellow slot ("Attention" or "Offline") to see the module issues.
Power Module Status Cage N	In a stacked system: each cage's own slots, with its own Alarm 1 and Alarm 2. Cage 1 is the cage that holds the system values shown in the System fields.

Transfer switch fields

Field	Description
System Output (V ac)	The output voltage of the connected power source. It should be close to the standard AC voltage of the region: 120 or 240 VAC.
Power Switch Temp (C)	The temperature of the power switch, in °C.
Utility Voltage	The AC voltage provided by the local utility. It should be close to 120 VAC.
Utility Current	The AC current provided by the local utility.
Utility Frequency	The AC frequency provided by the local utility. It should be close to 60 Hz.
Source	The current power source of the system: "Inverter" or "Utility".
Inverter	"Good" when the inverter source is active and working, otherwise "Fail".
Utility	"Good" when the utility source is active and working, otherwise "Fail".
Sync	"Active" when phase synchronization matches the inverter's output with the utility's, "None" when it does not.
Power Switch	"Normal" when the power switch temperature is within range, otherwise "Hot".
MBS	The maintenance bypass switch: "Inactive" when it is not in use, otherwise "Active".
Backfeed	Backfeed protection for the utility: "Inactive" while it has not acted, otherwise "Active".

Field	Description
Primary	The main power source chosen by the primary select switch: "Inverter" or "Utility".

3.10 Notification Message Box

The notification message box appears near the top of the tabbed pane to tell you about something you need to know. For example, it appears when the system has more power modules than are checked in System Configuration (Figure 3.18).



Figure 3.18. Notification Message Example

4. Running StatusBoost

4.1 Introduction

This section describes how to connect StatusBoost to your system so that it shows the latest data.

4.2 Procedure to Run Software

1. Select "Device Connection" in the "Tools" menu. The Device Connection window appears (Figure 4.1).

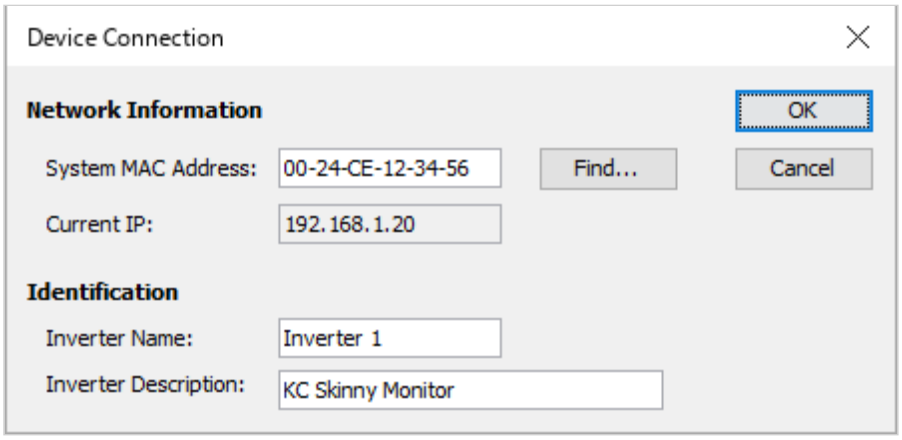


Figure 4.1. Device Connection Window

2. Enter the system's MAC address in the "System MAC Address" field and click "OK". To fill the field in automatically, click "Find..." (Section 5.2.1). Section 5.2, "Device Connection", describes the window.
3. The status field now shows "Stopped" instead of "Disconnected", which means StatusBoost is ready to run.
4. Select "Run SB" in the "Run" menu, or click the green triangle on the toolbar.

The status field now shows "Running", and the data fields fill with the system's data (Figure 4.2). To stop StatusBoost, select "Stop SB" in the "Run" menu or click the red square on the toolbar.

Messages you may see:

- **"MAC Address Error"** again: StatusBoost did not find a system with that MAC address on the network, or you clicked "Run SB" in a tab without a system. Check the address, and that the system is on and connected.
- **"Device IP found!"**: you left the IP address empty, and StatusBoost found it.
- **"Unknown Device"** when you open a "Tools" item: StatusBoost has not reached the system yet, so it doesn't know what kind of device it is. As the message says, check the details in the Device Connection window and click "OK".



Figure 4.2. StatusBoost Running Example

4.3 Setting Up a Stacked System

A **stacked system** is one inverter system made of several devices that StatusBoost shows together:

- up to 5 KC cages, connected to each other over CAN; and
- optionally the STS controller that is their transfer switch.

Each cage has its own network connection. Cage 1 holds the system values, which appear in the System fields.

To set up a stacked system:

1. Select "Device Connection" in the "Tools" menu and click "Find...".
2. In the Find Device window, check "Include legacy devices" if the system has an STS controller, then click "Refresh". The STS controller is listed only with that box checked.
3. Check the **Stack** box of every device of the system, then click "OK" (Figure 4.3).

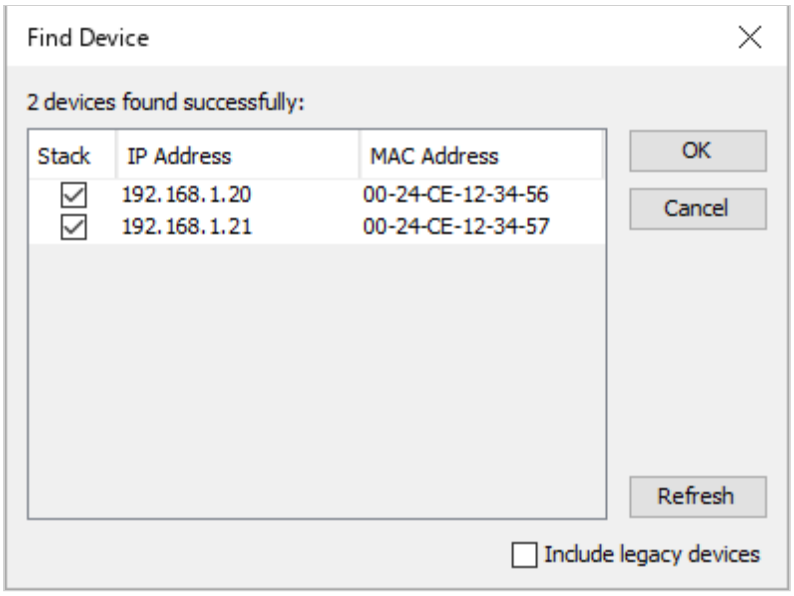


Figure 4.3. Devices Checked for a Stacked System

4. StatusBoost asks you to confirm (Figure 4.4). Click "Yes".

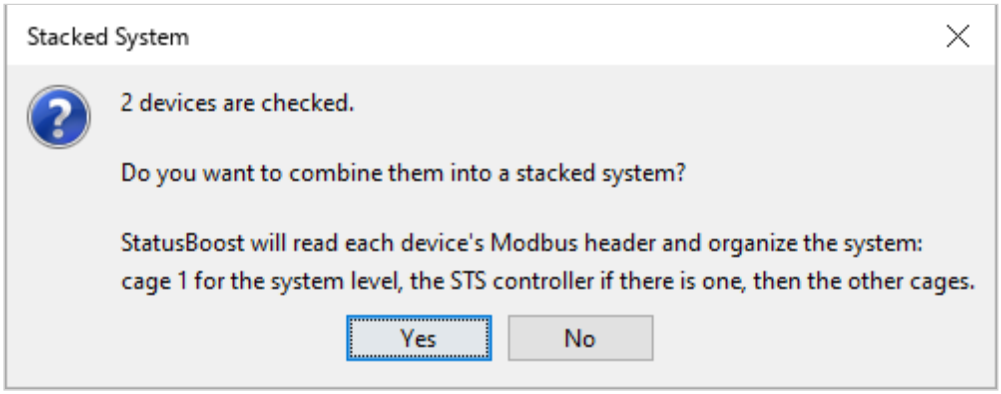


Figure 4.4. Confirming a Stacked System

5. StatusBoost reads each device to find out what it is, and the cage number of each KC cage. It then shows how it organized the system (Figure 4.5). Click "OK".

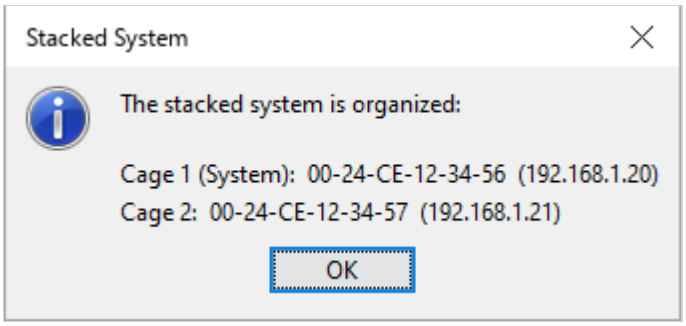


Figure 4.5. The Organized Stacked System

6. Back in the Device Connection window (Figure 4.6), cage 1 is filled in and the name is "Stacked System". Change the name and description if you like, then click "OK".

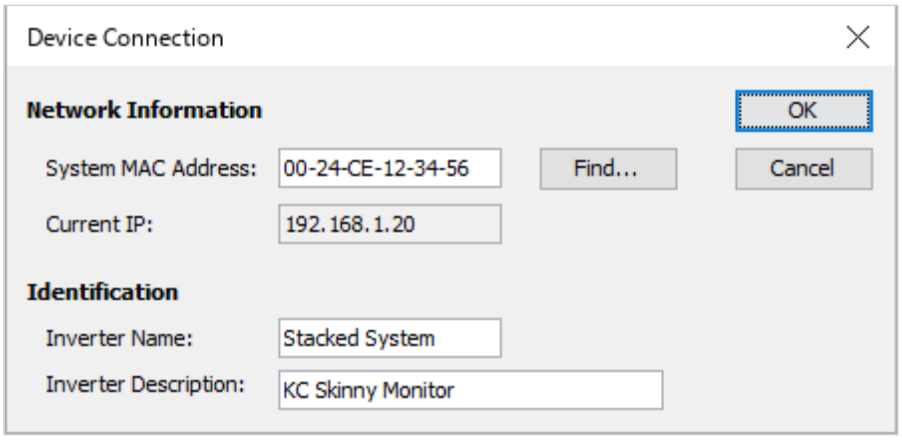


Figure 4.6. Device Connection with a Stacked System

7. Run StatusBoost as in Section 4.2. The window shows each cage (Figure 3.17): - System shows cage 1's values. - Power Module Status Cage 1 shows cage 1's modules and alarms. - Power Module Status Cage 2 and further sections show each further cage's modules and alarms. - With an STS controller in the system, the transfer switch fields are shown too.

StatusBoost saves the stacked system in "statb.ini" and shows it again the next time it starts. It also saves the address it found the STS controller at; if the controller stops answering there, StatusBoost forgets that address and looks for the controller by its MAC address.

With one Stack box checked, or none, "OK" chooses that device, or the selected one, as a single system.

StatusBoost does not build the stacked system, and says why, when:

- more than 6 devices are checked;
- more than 5 KC cages, or more than one STS controller, are checked;
- no KC cage is checked;
- a checked device is a KC transfer switch. It is a system of its own and is not stacked;
- a checked device is neither a KC cage nor an STS controller;
- none of the checked cages is cage 1, two cages report the same cage number, or a cage number is not between 1 and 6. Check the cages' CAN settings.

To go back to a single system, open Find Device and click "Refresh". Then select one device without checking any Stack box, and click "OK" in both windows. Renaming a stacked system in the Device Connection window keeps it stacked.

The menus of a stacked system. Network Feature Setup and System Configuration set up one networked device, so StatusBoost hides them while it shows a stacked system. The "Tools" menu then has only "Device Connection" (Figure 4.7).

- In the factory profile, "Energy Calibration", "AC/DC Configuration", "Rectifier Settings" and Factory > "Toggle Reference" are hidden too, and Rectifier Settings does not open when StatusBoost starts. "Assign System Model & Serial" stays: it finds its unit by the barcode, whatever system is shown.
- The items come back as soon as StatusBoost shows a single device again.

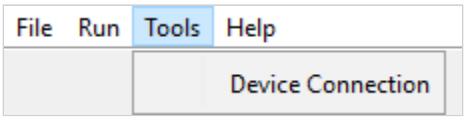


Figure 4.7. Tools Menu of a Stacked System

To configure one cage of a stacked system:

1. Go back to a single system with that cage, as described above.
2. Configure the cage with the "Tools" menu (Section 5).
3. Set up the stacked system again (steps 1 to 6 above).

4.4 Connecting to a Standalone STS Controller

An **STS controller** is the controller of a static transfer switch. It switches the output between the utility and the inverter. StatusBoost can show an STS controller on its own, or as the transfer switch of a stacked system (Section 4.3).

To connect to an STS controller:

1. Select "Device Connection" in the "Tools" menu and click "Find...".
2. In the Find Device window, check "Include legacy devices", then click "Refresh". An STS controller answers the search as a legacy device, so it is listed only with that box checked.
3. Select the STS controller without checking its Stack box, and click "OK" in both windows. You can also enter its MAC address in the Device Connection window instead.
4. Run StatusBoost as in Section 4.2.

StatusBoost recognizes the STS controller and shows its own data instead of the KC fields (Figure 4.8, Table 4.1). The status field is in the upper-right corner of the STS Controller section, with the same states as for a KC system (Section 3.8).

StatusBoost saves the address it found the controller at. If the controller stops answering there, StatusBoost forgets that address and looks for the controller by its MAC address.

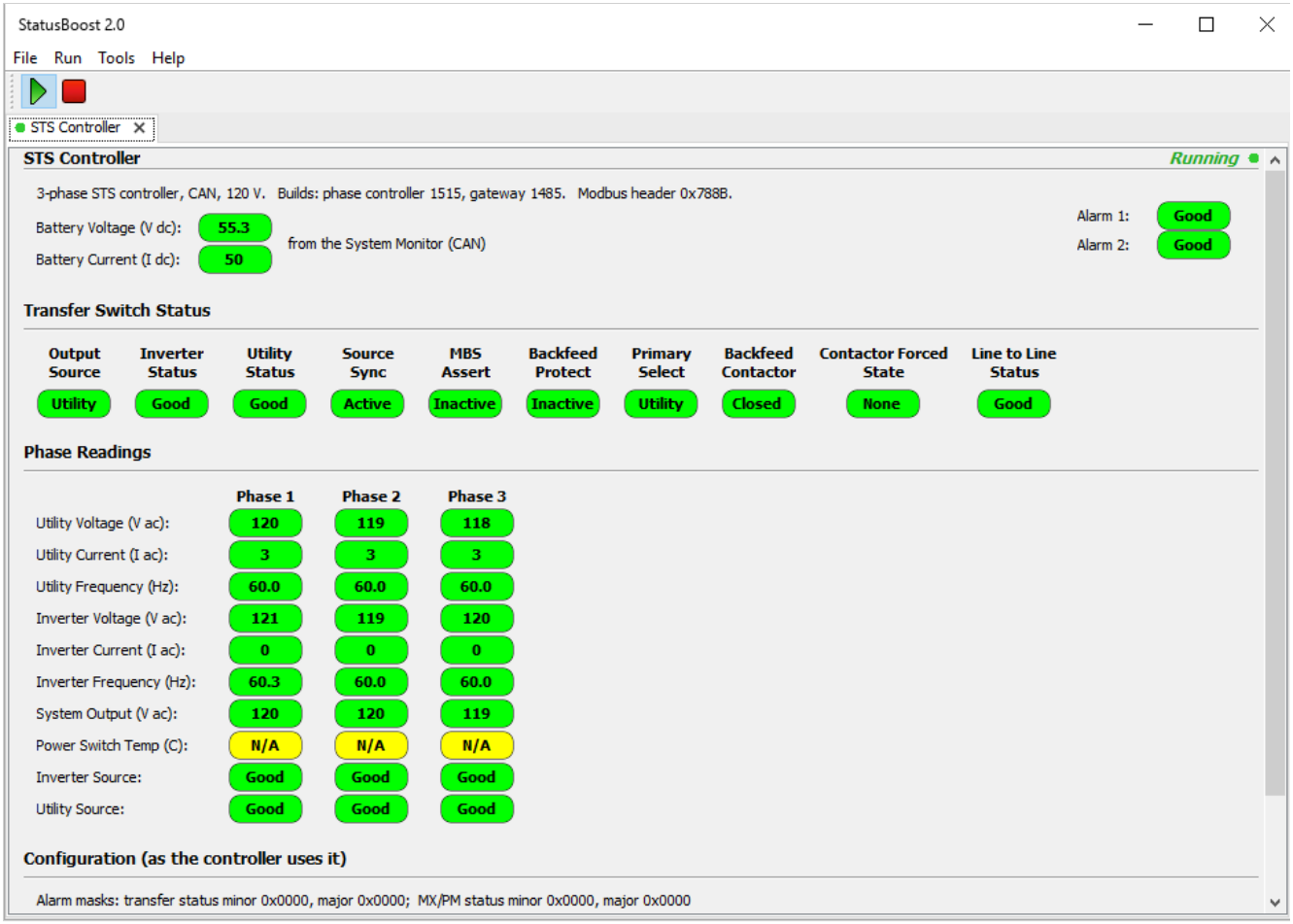


Figure 4.8. A Standalone STS Controller Running

Table 4.1

Field	Description
STS Controller	The first line describes the controller: its number of phases, how it reaches its phases (CAN or I2C), its voltage (120 V or 277 V, and "RIC" for a RIC unit), its phase controller and gateway firmware builds, and its Modbus header.
Battery Voltage, Battery Current	Shown only when a System Monitor supplies them to the controller, over CAN or I2C. A controller without one has no battery readings, and the fields are not shown.
Alarm 1 (Minor level)	"Good", or "Warning" while the controller reports a minor alarm.
Alarm 2 (Major level)	"Good", or "Fail" while the controller reports a major alarm.
Output Source	The source that feeds the output: "Utility" or "Inverter".
Inverter Status, Utility Status	"Good", or "Fail" while the controller rates that source bad.
Source Sync	"Active" while the two sources are in sync, "None" while they are not.
MBS Assert	"Active" while the maintenance bypass switch locks out transfers, otherwise "Inactive".

Field	Description
Backfeed Protect	"Active" (red) while backfeed protection is active, otherwise "Inactive".
Primary Select	The preferred source: "Utility" or "Inverter".
Backfeed Contactor	"Closed", or "Open" when the controller detects that the contactor is open.
Contactor Forced State	"None", or the source the backfeed sequence forced the contactor to: "Inverter" or "Utility".
Line to Line Status	Only for a controller with more than one phase: "Good", or "Fail" on a line-to-line fault.
Phase Readings	One column for each phase: the utility, inverter and output values, the power switch temperature, and the state of each source.
Power Switch Temp (C)	The temperature at the phase's power switch. "N/A" when no temperature sensor is connected, or the sensor is open.
Inverter Source, Utility Source	"Good", or what the controller found wrong with that source on that phase: "Voltage", "Freq" (frequency), "Fast" (a fast voltage failure) or "Fail". Point at the field to see the controller's failure code. The field shows "Good" again once the source has recovered.
Configuration (as the controller uses it)	The settings the controller works with: its alarm masks, network settings, SNMP DC voltage thresholds and calibration trims, and whether the settings' checksum (CRC) matches them. Scroll down to see them.

The menus with an STS controller. StatusBoost shows an STS controller's settings but does not change them yet. So "Network Feature Setup" and "System Configuration" are unavailable (greyed out). In the factory profile, "Energy Calibration", "AC/DC Configuration", "Rectifier Settings" and Factory > "Toggle Reference" are unavailable too, and Rectifier Settings does not open when StatusBoost starts. "Device Connection" and "Assign System Model & Serial" remain available.

4.5 Monitoring Several Systems

StatusBoost can monitor up to 6 systems at once, each in its own tab (Figure 4.9). A tab can hold a single system, a stacked system (Section 4.3) or a standalone STS controller (Section 4.4). Every tab is read at the same time, also while another tab is shown.



Figure 4.9. Three Systems in Their Tabs

To add a system:

1. Select "New Tab" in the "File" menu, or press Ctrl+T. A new tab opens, and the Device Connection window with it.
2. Choose the system as in Section 4.2, 4.3 or 4.4, and click "OK". If you click "Cancel", the new tab closes again.
3. Run it with "Run SB", or every tab with "Run All" in the "Run" menu.

"New Tab" is unavailable while 6 tabs are open.

The menus and the toolbar act on the selected tab. "Device Connection", the other "Tools" items, "Run SB", "Stop SB" and the toolbar buttons work on the system of the tab you see. "Run All" and "Stop All" in the "Run" menu work on every tab.

One device, one tab. A device can be in one tab only.

- The Find Device window does not list the devices of the other tabs. The line above the list says how many it left out, for example "3 devices found successfully (2 more are in other tabs):".
- If you enter the MAC address of a device that another tab has, StatusBoost says which tab has it.

To delete a tab, click the cross on its header, or select "Delete Tab" in the "File" menu for the selected tab.

- StatusBoost asks first (unless the tab has no system yet), then stops reading the tab's system and removes it from "statb.ini". The units themselves are not changed.
- When you delete the last tab, an empty tab remains.

When StatusBoost starts, it opens the tabs of the last session, with the tab you had selected. They open stopped: use "Run All" in the "Run" menu to start them.

A tab without a system, such as the first tab the first time, is used by "New Tab" before a new tab is added.

A message from a system, for example the Device Mismatch question, brings its tab to the front. With several tabs, the message's title starts with the system's name.

4.6 Monitoring a UC Alarm Card (StatusBoost .NET Only)

A **UC alarm card** (also called the LCNC card) monitors a UC inverter system: its battery, its inverters and its power module slots. StatusBoost .NET shows a UC card on its own, or several UC cards with an STS controller as a stacked system. StatusBoost, the Java edition, does not show UC cards.

To connect to a UC card:

1. Select "Device Connection" in the "Tools" menu and click "Find...".
2. In the Find Device window, check "Include legacy devices", then click "Refresh". A UC card answers the search as a legacy device, as an STS controller does (Section 4.4).
3. Select the UC card and click "OK" in both windows. You can also enter its MAC address in the Device Connection window instead.
4. Run StatusBoost as in Section 4.2.

StatusBoost .NET recognizes the UC card and shows it in the usual sections (Figure 4.10, Table 4.2).



Figure 4.10. A UC Alarm Card Running (StatusBoost .NET)

Table 4.2

Field	Description
Battery Voltage, Battery Current	The battery's DC voltage and current.
Inverter Voltage, Current, Frequency	One column for each phase the card monitors.
AB Bus	Shown while the card monitors the AB bus.
Alarm 1, Alarm 2	Alarm 1 is the card's minor alarm, Alarm 2 its major alarm: "Good", or the alarm's state.
Power Module Status	Always all six slots. Each shows "OK", or "I2C Error" when the card could not read that slot. A UC card has no module details: no model, serial number or state, and double-clicking a slot shows nothing.
Utility values, System Output, Power Switch Temp, Transfer Switch Status	Only when an STS controller answers on the card's CAN bus, as for a KC system with an STS controller.

Stacked UC cards. Up to five UC cards, CAN cages 1 to 5, and an STS controller as CAN cage 0 make a stacked system (Figure 4.11). Set it up as in Section 4.3, with "Include legacy devices" checked. A card's cage number is its cage ID switch; a switch at 0 counts as cage 1. Cage 1 shows the system values, and every further cage shows its six slots and its own Alarm 1 and Alarm 2. KC cages and UC cards can't be in one stacked system: only an STS controller joins either kind.

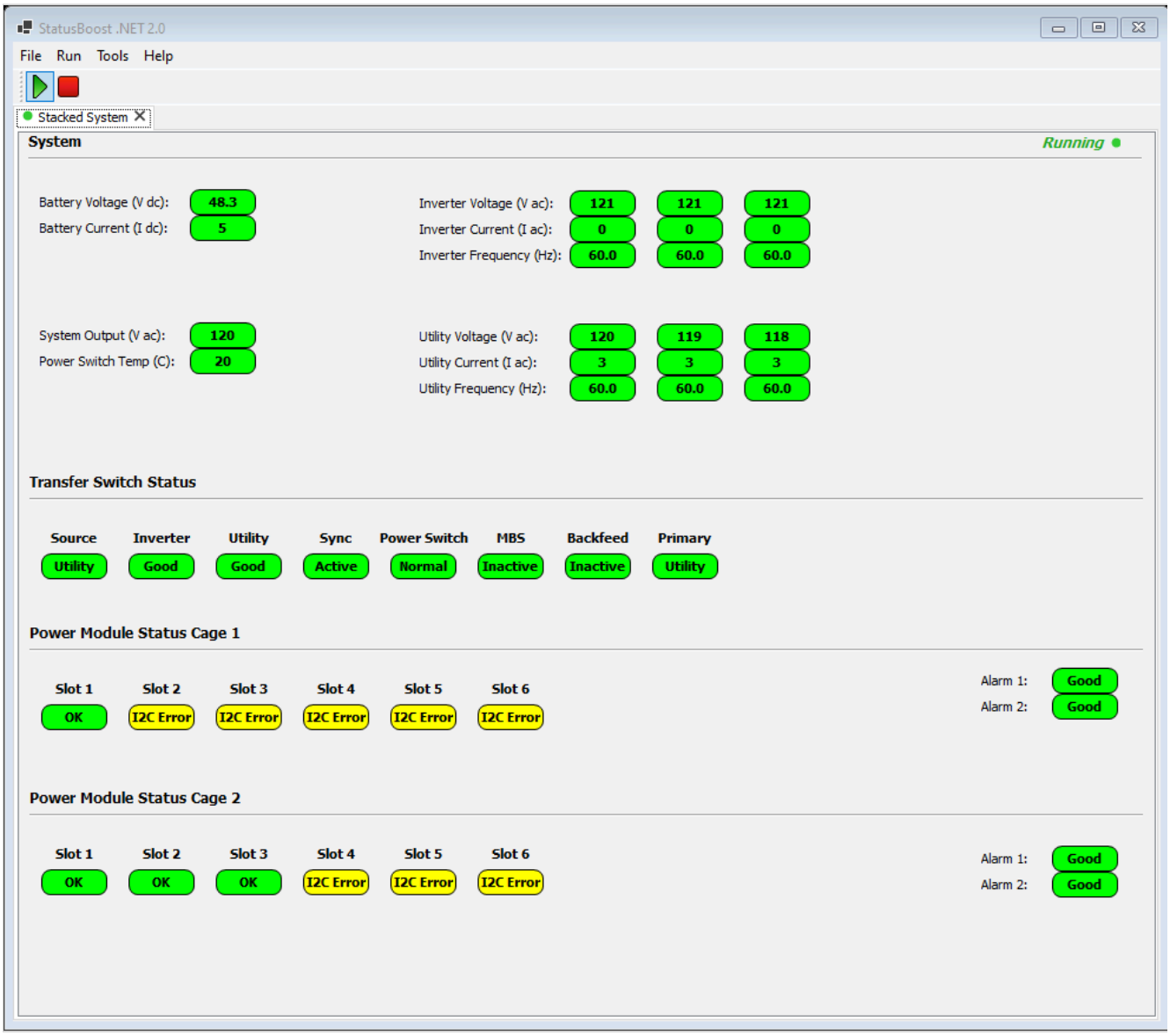


Figure 4.11. Two UC Cards and an STS Controller (StatusBoost .NET)

The menus with a UC card. StatusBoost .NET does not change a UC card's settings yet. So "Network Feature Setup", "System Configuration", "Energy Calibration", "AC/DC Configuration" and "Rectifier Settings" are unavailable (greyed out). Factory > "Toggle Reference" works on a UC card, as on a KC system (Section 3.5.3).

To try it without a card, start StatusBoost .NET with `--demo uc` (one simulated card) or `--demo ucstack` (two cards and an STS controller).

5. Tools Menu Selections

5.1 Introduction

Most of the options for configuring the inverter system are in the "Tools" menu.

While a configuration window is open, StatusBoost stops reading the tab's system, and starts again when you close the window if it was reading it before. Device Connection, System Configuration, Energy Calibration and AC/DC Configuration do this; the other tabs go on.

How configuration changes are sent. When you click "Update" in a configuration window, or "Set" in Rectifier Settings:

1. StatusBoost first reads the current settings from the system.
2. It sends only the settings you changed, each separately. A setting you did not change keeps the system's value, even if the window showed an older one.
3. It waits until the system confirms each one before it sends the next.

If you changed nothing, the window says "No changes detected." and sends nothing.

The system takes a few seconds to store a setting, so each changed setting can take about 10 seconds. The window waits until all the settings are confirmed.

If a setting is not confirmed, a message lists it, together with any settings that were not sent.

5.2 Device Connection

The Device Connection window is where you choose the system StatusBoost connects to, and how it is named (Figure 5.1).

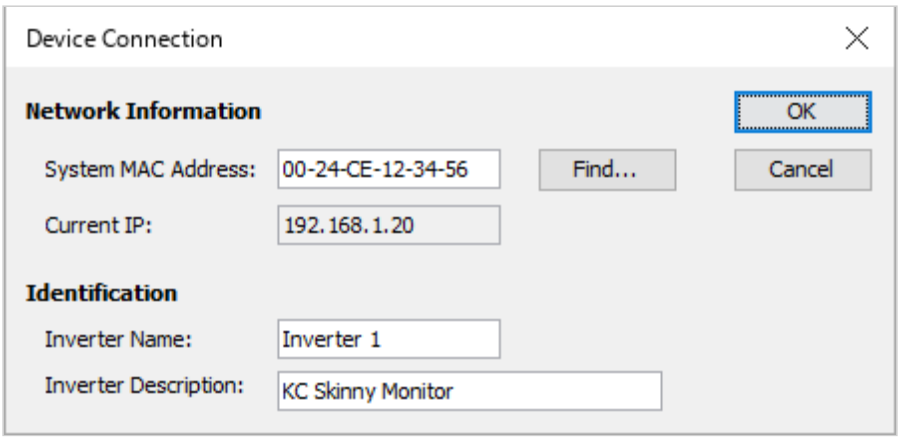


Figure 5.1. Device Connection Example

Field	Description
System MAC Address	The last six characters of the system's MAC address. The first six are the same for every Exeltech system. For a stacked system, this is cage 1.
Current IP	Read only. The system's current IP address, once the system has been found on the network.
Inverter Name	A name for your inverter system, shown on its tab. "Find..." fills in the device's MAC address, or "Stacked System" for a stacked system; you can change it. A system without a name is called "Inverter1".
Inverter Description	A description of your inverter system, for example its location in the building, how much power it delivers or its model type.

5.2.1 Find Device

Click "Find..." in the Device Connection window to open the Find Device window (Figure 5.2). It does not search when it opens: check "Include legacy devices" first if you need it, then click "Refresh". It does not list the devices of the other tabs (Section 4.5).

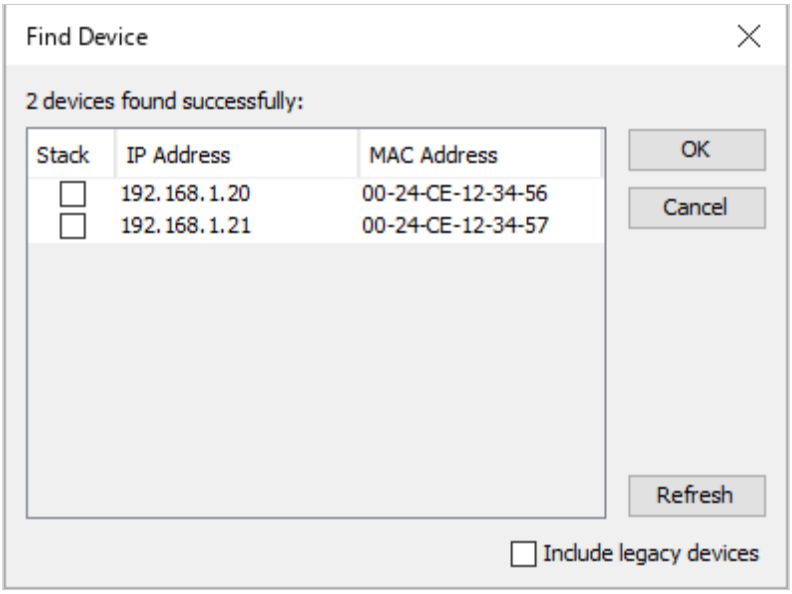


Figure 5.2. Find Device Example

Field	Description
Device List	Every device found on the network, with its IP and MAC address. For a single system, select the device and click "OK".
Stack	Check two or more devices to combine them into a stacked system (Section 4.3).
Refresh	Searches the network. Click it again to search again.
Include legacy devices	Also lists older Exeltech network devices. Check it before clicking "Refresh" to find an STS controller, or with StatusBoost .NET a UC alarm card: they answer as legacy devices.

Find Device searches the local network only.

5.3 Network Feature Setup

In the Network Feature Setup window you can set up the system's IP parameters, trap parameters for SNMP trap logging, and SNTP time synchronization with a local time server (Figure 5.3). It works with a single networked device; a stacked system hides it (Section 4.3). It is not available for an STS controller (Section 4.4).

The window opens at once and shows "Reading network settings from ..." until it has read the system's settings; the fields can be edited after that. If StatusBoost has no valid IP address for the system, the window says so and reads nothing. Connect to the system first (Section 5.2).

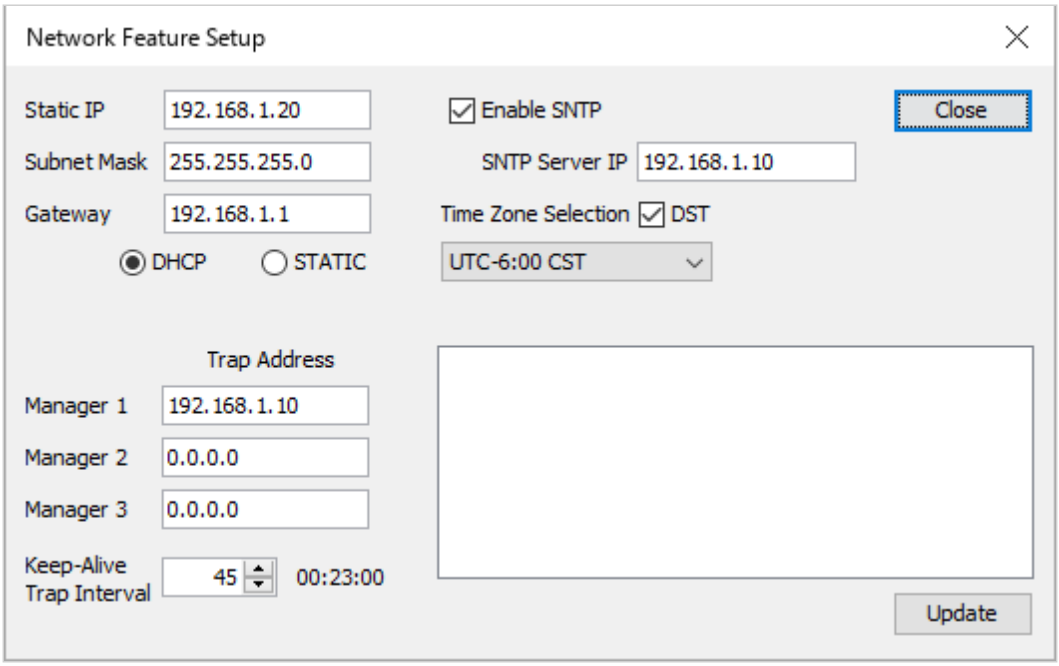


Figure 5.3. Network Feature Setup Example

Field	Description
IP Parameters	The static IP, subnet mask, gateway and IP address type (DHCP or static). DHCP and static are one choice: one of them is always selected.
Trap Event Parameters	The trap IP addresses and the keep-alive interval.
SNTP Parameters	Check "Enable SNTP" to use SNTP. Then enter the SNTP server IP (the local time server) and select its time zone. Check "DST" to take daylight saving time into account.
Status Log	The large text box at the lower right shows the result of every parameter update.

"Update" first checks the addresses. A subnet mask must be a valid mask, the static IP must be a host address in its subnet, and the gateway must be another host in the same subnet. A trap manager is 0.0.0.0 (not used) or a host address, and with SNTP enabled the SNTP server must be a host address. A field that fails a check turns red, and the Status Log says why.

Then StatusBoost lists the changes and asks you to confirm, because the network card restarts to use them. "Cancel" sends nothing. When the static addresses or the IP address type change to static, it also warns you: if the addresses are wrong, the system can no longer be reached over the network.

"Update" sends only the settings you changed, then restarts the system's network card ("Network Rebooting..."):

- The system's SNMP agent and web server stay as they were.
- The static IP, subnet mask and gateway are sent together, in one request.
- If nothing was changed, nothing is sent ("No Changes to Update").
- If a setting could not be sent, the network card is not restarted. Click "Update" again to send what is still missing.
- If StatusBoost could not read a setting when the window opened, the field stays blank, the Status Log lists it, and "Update" is disabled. Close the window and open it again to retry.

The Static IP jumper. The monitor card of a KC Skinny system has a Static IP jumper. It is used only when a hard reset of the network card is initiated with the hard reset button, which every networked Exeltech device has: with the jumper fitted, the network card starts with a static IP address; with it open, it uses DHCP. At all other times the IP address type is the one set in this window.

5.4 System Configuration

In the System Configuration window you set up the alarms, and which module slots and cages the system has (Figure 5.4). The changes are sent as described in Section 5.1. It works with a single networked device; a stacked system hides it (Section 4.3). It is not available for an STS controller (Section 4.4).

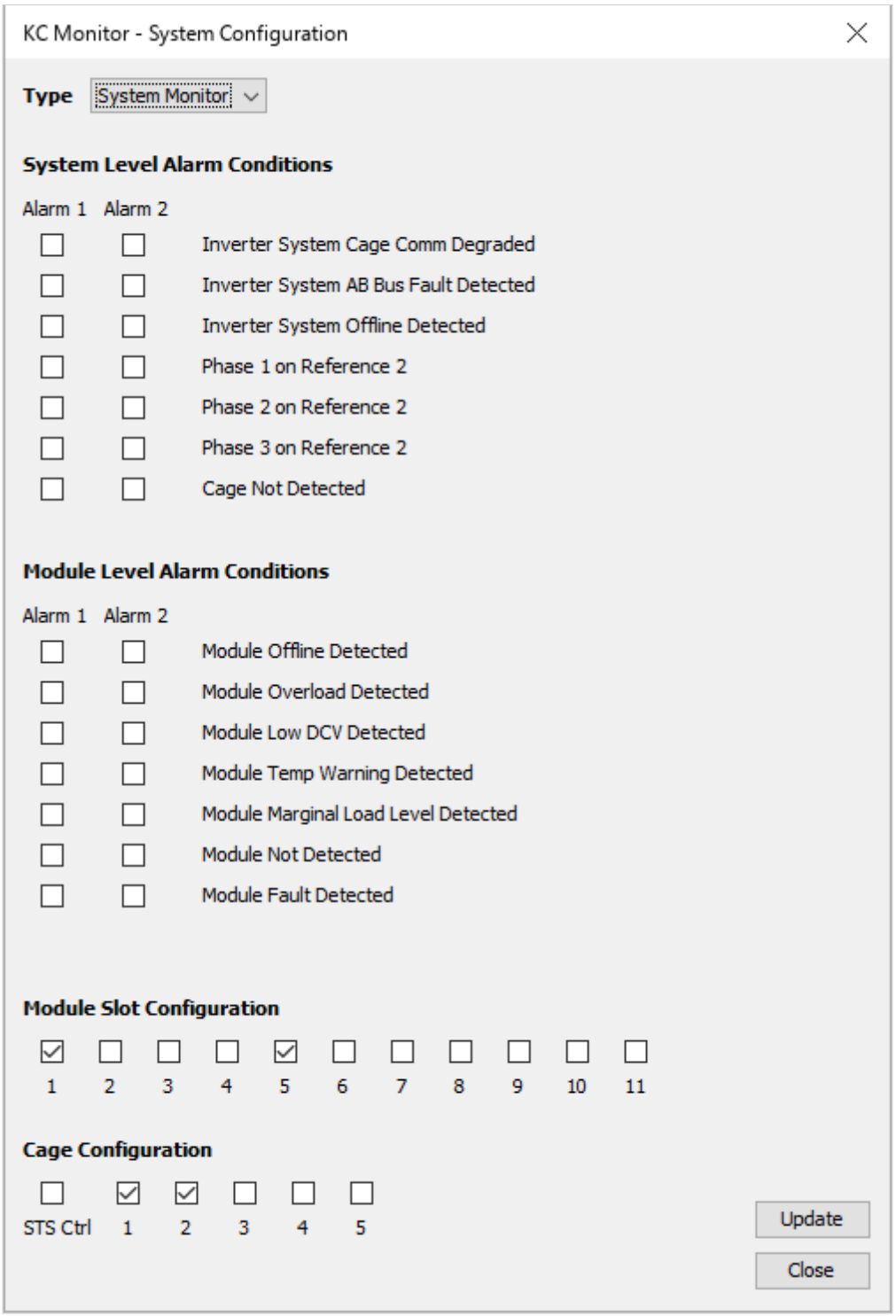


Figure 5.4. System Configuration Example

Field	Description
Type	"System Monitor" (a system with a monitor card only) or "STS Controller" (a system with a monitor card and a static transfer switch: a KC transfer switch, not the standalone STS controller of Section 4.4).

Field	Description
Alarm Conditions	The conditions that raise Alarm 1 (minor level) or Alarm 2 (major level).
Module Slot Configuration	The module slots shown in the main window. A checked slot with no module shows "N/A".
Cage Configuration	The cages the system is expected to have. To raise an alarm for a cage that is not detected, also check one or both "Cage Not Detected" alarm conditions. This depends on the system's firmware supporting cage detection.

5.5 Energy Calibration (Factory Profile Menu Only)

In the Energy Calibration window you calibrate the phase voltages and currents (Figure 5.5). A stacked system hides it (Section 4.3).

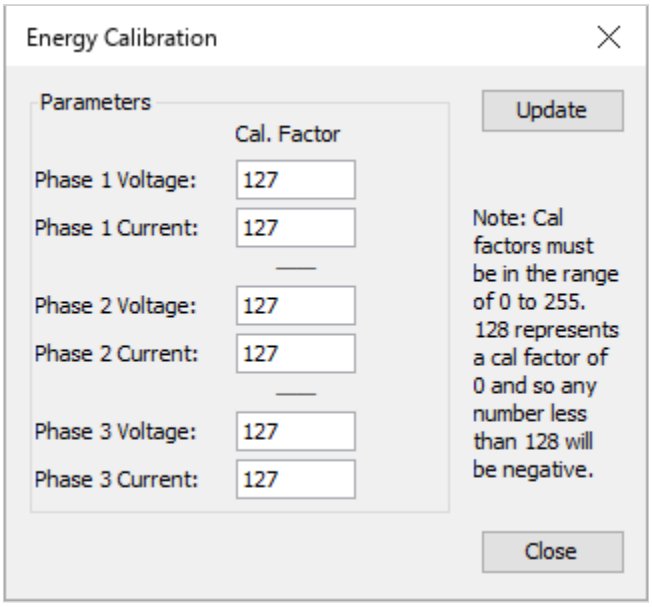


Figure 5.5. Energy Calibration Example

Field	Description
Phase Voltage Fields	A voltage calibration from 0 to 255. 128 represents 0, so smaller values are negative (127 = -1, 126 = -2, and so on).
Phase Current Fields	A current calibration from 0 to 255, with 128 representing 0 as above.

If a field is empty or not a whole number, StatusBoost says so and sends nothing.

5.6 AC/DC Configuration (Factory Profile Menu Only)

In the AC/DC Configuration window you set the system's AC and DC voltage and current parameters (Figure 5.6). A stacked system hides it (Section 4.3).

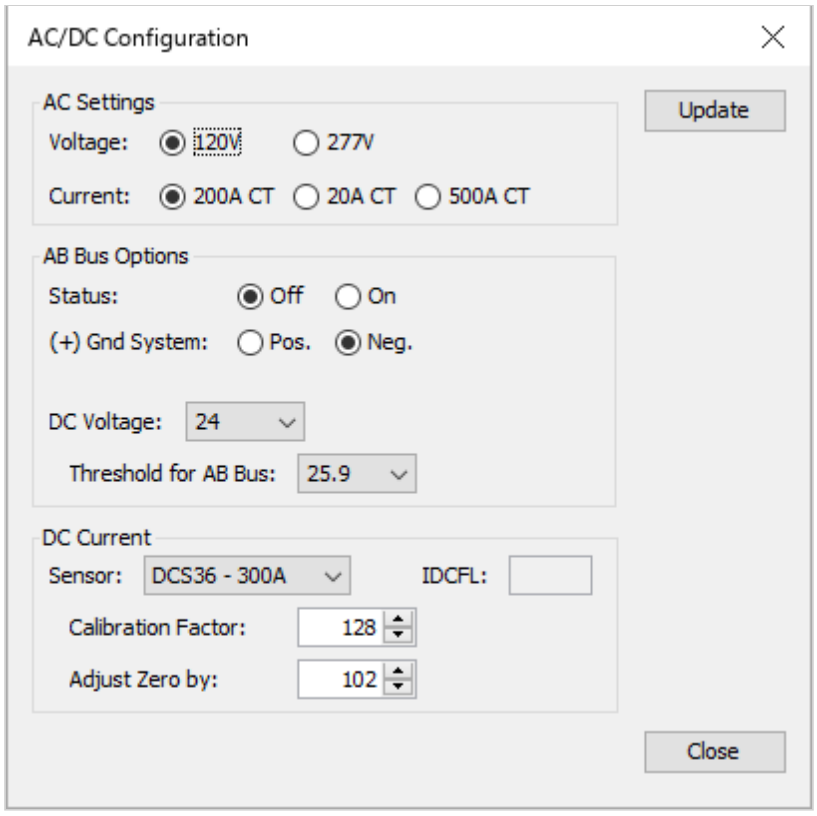


Figure 5.6. AC/DC Configuration Example

Field	Description
AC Voltage	The system's output voltage: 120 or 277 V.
AC Current	The current transformer: 200, 20 or 500 A.
AB Bus Status	"Off" when the system has no AB bus to monitor, "On" when it does. With "On", the main window shows the AB Bus field.
AB Bus Ground System	A positive or negative ground system for the AB bus.
DC Voltage	The system's nominal input voltage: 24, 48, 108 or 120 V.
Threshold for AB Bus	The AB-bus voltage threshold, chosen to suit the DC voltage.
DC Current Sensor	The Hall-effect sensor type and amp rating fitted to the system.
IDCFL	Read only. The latest suggested value for "Adjust Zero by".
Calibration Factor	A DC current calibration from -127 to 127, for the selected sensor.
Adjust Zero by	A zero adjustment for the DC current, from 0 to 519, for the selected sensor.

5.7 Rectifier Settings (Factory Profile Menu Only)

The Rectifier Settings window shows the rectifier's settings and their actual values, and sends new settings (Figure 5.7). In the factory profile it also opens when StatusBoost starts. A stacked system hides it, and it does not open at start for a stacked system (Section 4.3).

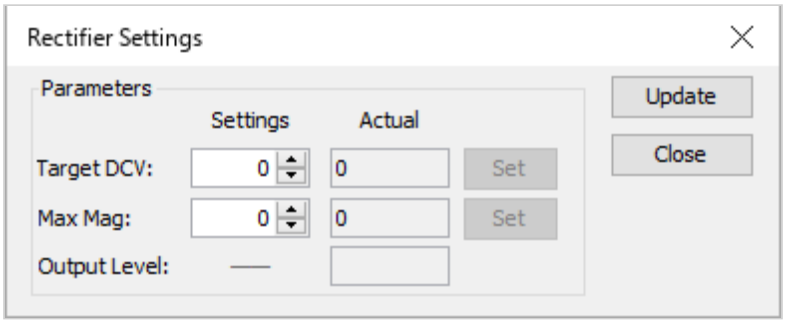


Figure 5.7. Rectifier Settings Example

Field	Description
Target DCV	The rectifier's target DC voltage, from 0 to 1023. It is sent only when you change it. The system cannot show its target, so "Actual" shows "unknown" when the system holds a target other than the one shown.
Max Mag	The rectifier's maximum magnitude, from 0 to 255.
Output Level	The rectifier's present output level (actual value only).

If the system does not confirm a new target DCV, StatusBoost says so and does not send Max Mag.

6. FAQs

6.1 Introduction

This section answers frequently asked questions about installing and using StatusBoost.

6.2 General Questions

Why doesn't StatusBoost start?

StatusBoost brings its own Java, so a missing or old Java on your PC does not matter.

1. **Start it from the Start menu** (in the "Exeltech" folder) or from the desktop shortcut.
2. **If it still doesn't start,** uninstall it (Section 2.5) and install it again (Section 2.3). Your settings are kept.
3. **To see the messages StatusBoost prints,** for example for Exeltech technical support, start it from a command prompt with its own Java. If you installed it in another folder, use that folder's name:

   ```
   "C:\Program Files\StatusBoost\runtime\bin\java.exe" -jar "C:\Program
   Files\StatusBoost\app\StatusBoost.jar"
   ```

   The messages then appear in the command prompt.

Do we have to keep the computer turned on for StatusBoost?

Yes. StatusBoost shows the system's status only while it is running.

StatusBoost shows "Disconnected", but the system is on. Why?

StatusBoost cannot reach the system over the network. Check:

- the network cable and switch;
- that the PC and the system are on the same network;
- that no firewall blocks TCP port 502.

StatusBoost keeps trying and reconnects by itself as soon as the system answers.

The system also accepts only two network connections at a time. If other programs read it at the same time, for example another copy of StatusBoost, close them.

Can I start StatusBoost twice to watch two systems?

Use tabs instead (Section 4.5): one StatusBoost watches up to 6 systems. Two copies of StatusBoost would each save their own systems in "statb.ini", and the last one to save would overwrite the other's.

6.3 Software Settings

Why doesn't StatusBoost remember my settings after I exit and start it again?

StatusBoost saves its settings in the "statb.ini" file in your settings folder, "%APPDATA%\StatusBoost". To open that folder, type `%APPDATA%\StatusBoost` in the File Explorer address bar. Upgrading or uninstalling StatusBoost leaves it in place. Section 2.4 explains how to bring over the settings of an earlier version. If the file is deleted or moved, StatusBoost creates a new "statb.ini" with default settings, and those are read the next time it starts.

With several tabs, "statb.ini" holds the systems of every tab. A version of StatusBoost older than 2.0.5 reads only the first tab's system, and forgets the others when it saves.

Where is my stacked system stored?

In "statb.ini": one section per device, "[Device]" for cage 1 and "[Device2]" to "[Device6]" for the others. The systems of the second to sixth tab have sections "[System2.Device]" to "[System6.Device6]", and "[Systems]" remembers the selected tab.

7. Troubleshooting

7.1 Introduction

This section covers situations that may arise while running StatusBoost and need troubleshooting.

7.2 Device Mismatch

When StatusBoost starts, it may show a "Device Mismatch" message (Figure 7.1). This means that the keys stored in the INI file do not match the keys read from your inverter system. The message names the device and lists the keys that differ.

- Click "Update" to change the INI file to match the system.
- Click "Skip" to leave it as it is. You can then restart the system to see whether that clears an error in its data.

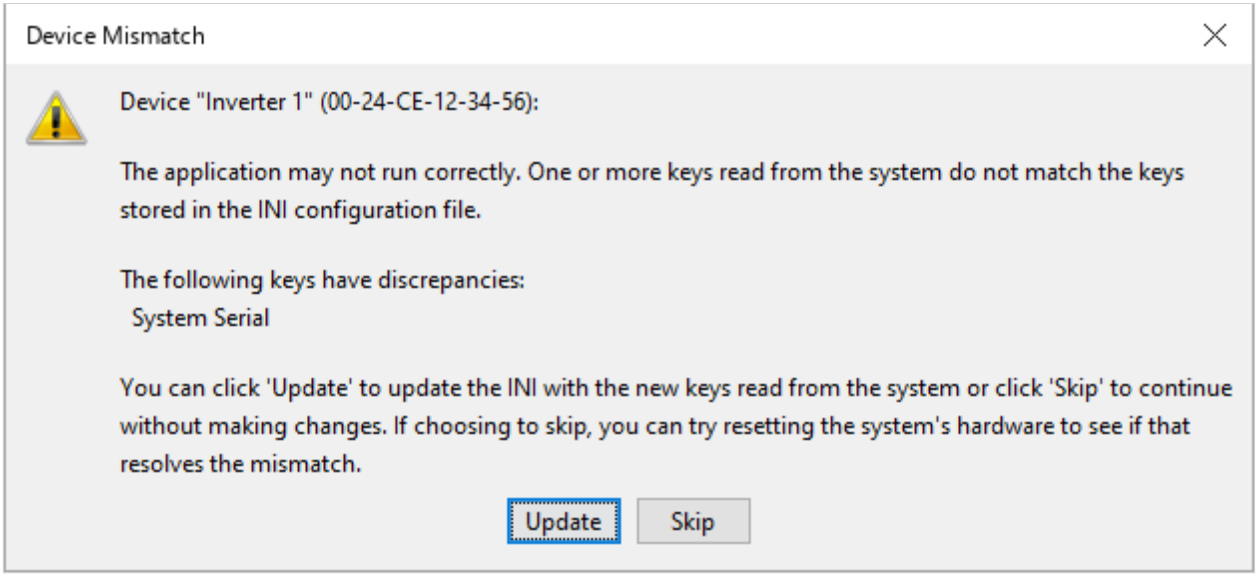


Figure 7.1. Device Mismatch

Reasons a mismatch can occur:

- The information wasn't saved to the INI file the last time StatusBoost ran.
- The INI file was edited before StatusBoost started.
- The data stored in the system was changed or reset before StatusBoost started.
- A different system now answers at the stored MAC address.

A device whose configuration was never stored in the INI file is not reported as a mismatch: StatusBoost stores its configuration.

7.3 A Stacked System Is Not Built

If StatusBoost does not build a stacked system, its message says why. Section 4.3 lists the reasons. The most common are:

- **Cage numbers.** Every KC cage reports its CAN cage number, and cage 1 must be among the checked devices. Two cages with the same number usually mean that they belong to different systems, or that a cage's CAN setting is wrong.
- **A device could not be read.** Make sure it is switched on and reachable, then try again.

A. File and Folder Specification

The installation folder, "C:\Program Files\StatusBoost" unless you chose another one (Section 2.3):

Table A.1

File Name	Description
StatusBoost.exe	Starts StatusBoost. The Start-menu entry and the desktop shortcut point to it.
app\StatusBoost.jar	The main StatusBoost program.
app\lib	The software libraries that StatusBoost uses.
app\sbguide.pdf	This StatusBoost User Guide, which "Help" → "User Guide" opens.
runtime	The Java that StatusBoost runs on. It is part of StatusBoost and not used by other programs.

Your settings folder, "%APPDATA%\StatusBoost" (Section 6.3):

Table A.2

File Name	Description
statb.ini	StatusBoost's settings: the devices of every tab (MAC address, Modbus key, system serial number, configuration in hex, name, description and phase count), the selected tab, the factory profile flag and the logs folder. Created when StatusBoost first starts; upgrading or uninstalling StatusBoost leaves it in place.

B. StatusBoost .NET

StatusBoost .NET is a second edition of StatusBoost, for Windows, with the same version numbers. It does what this guide describes, with these differences:

	StatusBoost	**StatusBoost .NET**
Window title	"StatusBoost 2.0"	"StatusBoost .NET 2.0"
Installer	"StatusBoost-<version>.msi"	"StatusBoost.NET-<version>-x64.msi"
Installation folder	"C:\Program Files\StatusBoost"	"C:\Program Files\StatusBoost .NET"
Start menu, desktop shortcut and Apps entry	"StatusBoost"	"StatusBoost .NET"
Windows Firewall rule	"StatusBoost"	"StatusBoost .NET"
Files (Appendix A)	"StatusBoost.exe", the "app" folder with "sbguide.pdf", and the "runtime" folder with its Java	"StatusBoost.exe" and "sbguide.pdf" only
Settings folder (Section 6.3)	"%APPDATA%\StatusBoost"	"%APPDATA%\StatusBoost.NET". At its first start, StatusBoost .NET copies "statb.ini" from "%APPDATA%\StatusBoost" if there is one; after that the two keep their own settings
About box	StatusBoost's version and the Java it runs on	StatusBoost's version and the .NET it runs on
Messages for technical support (Section 6.2)	Shown when StatusBoost is started from a command prompt	Not shown yet
UC (LCNC) alarm cards (Section 4.6)	Not shown	Shown, on their own or stacked with an STS controller

Both can be installed on the same PC. Don't let both read the same system at the same time: the system accepts only two network connections at a time (Section 6.2).